

Student Feedback Analysis System

Final Scope of Improvement Report

-by Pareet Mistry

1. Executive Summary

This report consolidates findings from a full-cycle analysis of 2,000 student feedback records collected across programs, batches, and institutions run by Ardent Software Services. The analysis combined descriptive statistics, correlation analysis, a classical machine-learning model, k-means clustering, and text/sentiment mining of open-ended responses to identify the specific, prioritized scope of improvement in training delivery.

Across four independent analytical methods, two dimensions consistently emerged as the organization's most significant improvement priorities:

- **Pace of Delivery** — rated lowest on average, highly inconsistent across students, a meaningful predictor of dissatisfaction, and the single most frequently cited complaint in open-ended feedback (132 mentions of feeling “rushed”).
- **Platform / Infrastructure** — the lowest-rated and most inconsistent dimension overall, with direct textual complaints about lag and the online/live delivery experience.

By contrast, Trainer Effectiveness and Support & Mentorship are the organization's clearest strengths — both are rated highly and are the strongest statistical drivers of overall satisfaction, meaning current investment in these areas should be protected rather than redirected.

2. Methodology

The analysis followed the six-week phased approach defined in the project requirement document:

- Phase 1 — Data cleaning: missing-value imputation (median for ratings), categorical normalization, duplicate and range validation.
- Phase 2 — Exploratory data analysis: descriptive statistics, segmentation by Program/Batch/Institution, and correlation analysis.
- Phase 3 — Classical ML modeling: a logistic regression model to identify predictors of low satisfaction, and k-means clustering to segment students into feedback profiles.
- Phase 4 — Qualitative analysis: lexicon-based sentiment scoring (VADER) and TF-IDF keyword extraction on open-ended responses.

All modeling used classical, interpretable methods only (logistic regression, k-means, TF-IDF, lexicon-based sentiment), consistent with the project's requirement that findings remain explainable to non-technical stakeholders. No deep learning or generative/LLM-based methods were used.

3. Quantitative Findings

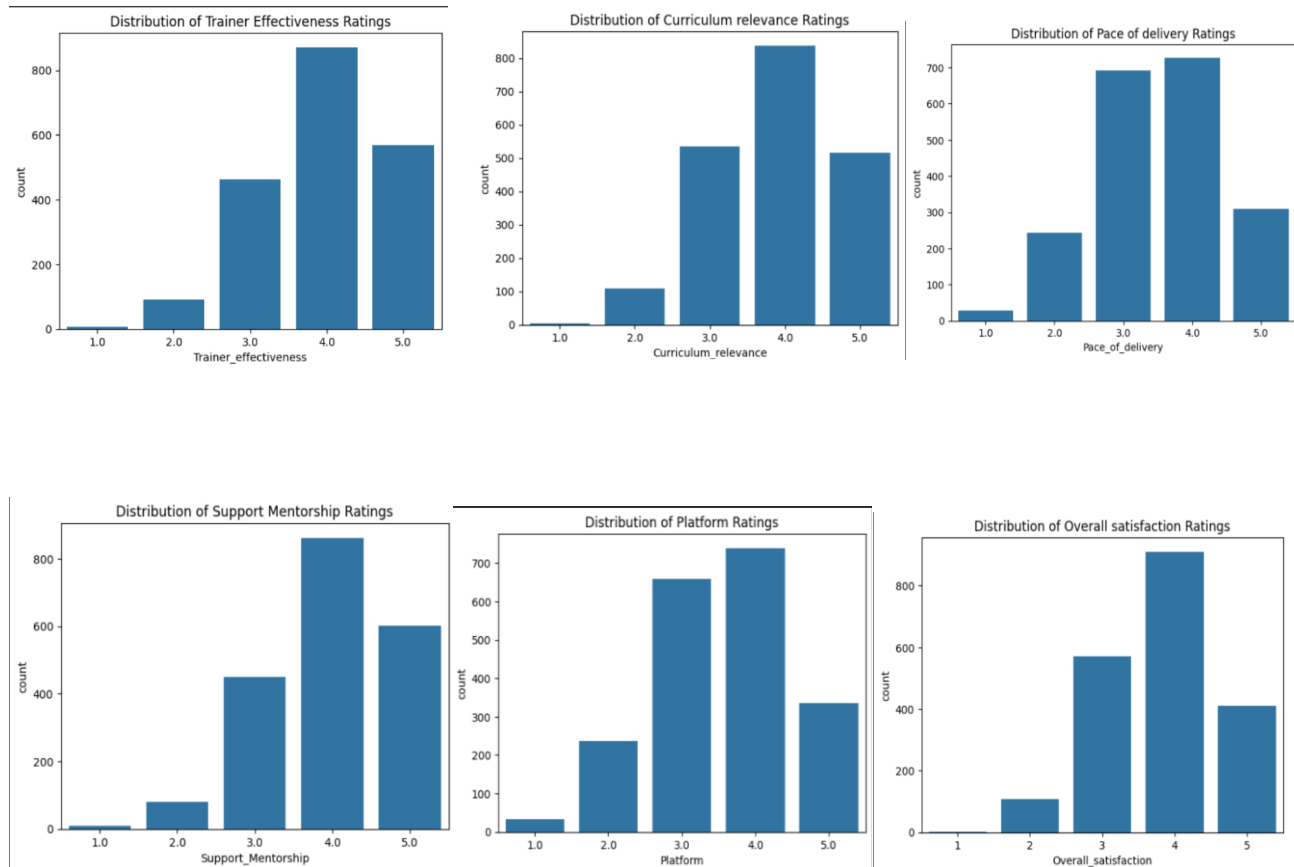
3.1 Descriptive Statistics & Correlation

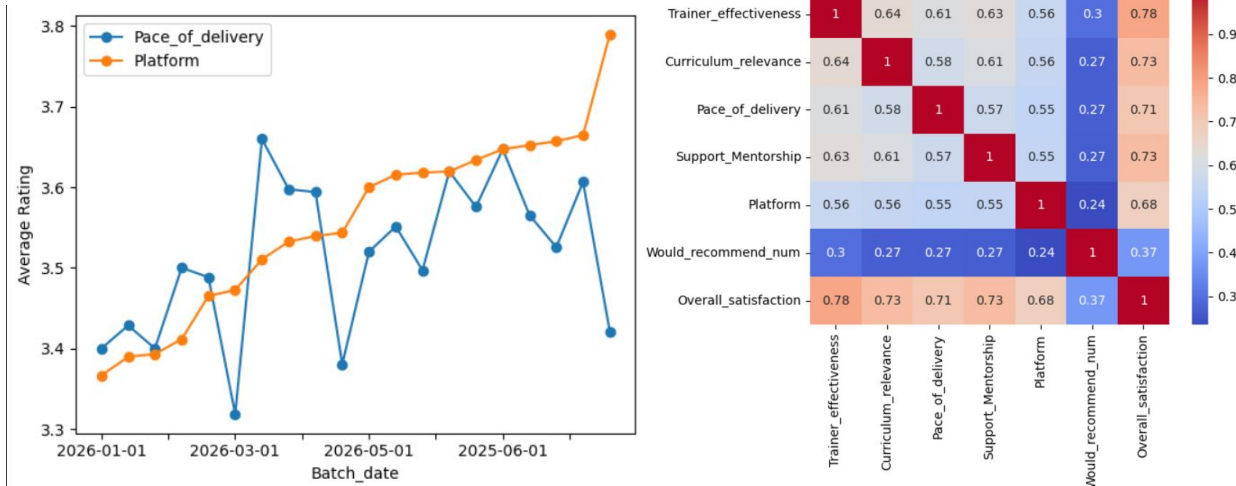
Mean rating, variability, and statistical relationship to Overall Satisfaction for each of the five feedback dimensions (1–5 scale):

	Trainer_effectiveness	Curriculum_relevance	Pace_of_delivery	Support_Mentorship	Platform	Overall_satisfaction
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	3.952500	3.87550	3.523500	3.984000	3.554500	3.811000
std	0.852412	0.86335	0.941218	0.849179	0.956288	0.820741
min	1.000000	1.00000	1.000000	1.000000	1.000000	1.000000
25%	3.000000	3.00000	3.000000	3.000000	3.000000	3.000000
50%	4.000000	4.00000	4.000000	4.000000	4.000000	4.000000
75%	5.000000	5.00000	4.000000	5.000000	4.000000	4.000000
max	5.000000	5.00000	5.000000	5.000000	5.000000	5.000000

Pace of Delivery and Platform show the lowest average scores and the highest variance, indicating both weaker and less consistent experiences. Trainer Effectiveness and Support & Mentorship show the strongest correlation with, and predictive influence on, Overall Satisfaction — confirming they are the highest-impact dimensions for the student experience, and are currently performing well.

OUTPUTS:





3.2 Predictive Modeling

A logistic regression model was trained to predict “low satisfaction” (defined as an Overall Satisfaction rating of 3 or below) from the five rating dimensions. The model achieved 88.6% accuracy on held-out test data (recall of 81% for identifying genuinely low-satisfaction students), meaningfully outperforming a majority-class baseline of ~64%. Trainer Effectiveness and Support & Mentorship were the strongest predictors; Curriculum Relevance was comparatively the weakest, once the overlap between dimensions was accounted for.

Accuracy score = 0.886				
	precision	recall	f1-score	support
0	0.90	0.93	0.91	319
1	0.86	0.81	0.84	181
accuracy			0.89	500
macro avg	0.88	0.87	0.87	500
weighted avg	0.89	0.89	0.89	500

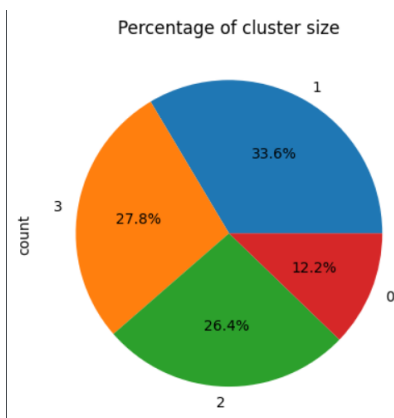
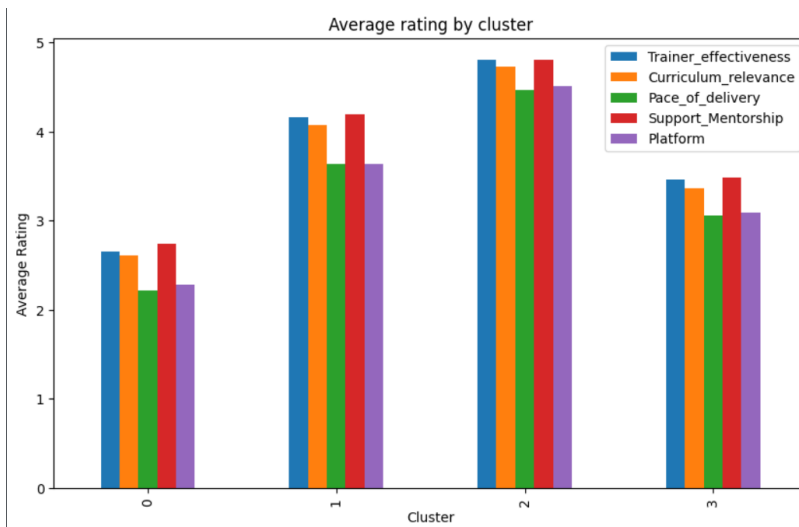
3.3 Student Segmentation (Clustering)

K-means clustering on the five rating dimensions identified four distinct student feedback profiles:

Cluster Profile	Avg. Rating Range	Interpretation
Highly Satisfied	4.5 – 4.8	Consistently high across all five dimensions; the largest positive reference group.

Cluster Profile	Avg. Rating Range	Interpretation
Moderately Satisfied	3.6 – 4.2	Solid overall, but Platform and Pace of Delivery trail behind the other dimensions.
Neutral / Mixed	3.1 – 3.5	Fairly flat scores across dimensions, no dimension stands out as a specific driver.
Dissatisfied	2.2 – 2.7	Lowest across every dimension, with Pace of Delivery and Platform the weakest even within this group; highest-priority group for direct outreach.

The Dissatisfied cluster represents the organization's highest-risk group for retention and requires broad, cross-dimensional intervention rather than a single fix; the Moderately Satisfied cluster's specific weak points (Platform, Pace) mirror the org-wide findings.



4. Qualitative Findings (Text & Sentiment Analysis)

Sentiment analysis of open-ended comments (1,949 responses) showed an overall positive skew, broadly consistent with the numeric ratings: 54% Positive, 33% Neutral, 10% Negative.

Keyword extraction (TF-IDF) on negative-sentiment comments surfaced terms directly corroborating the statistical findings: “pace”, “fast”, “moved”, “quickly” (delivery speed) and “platform”, “lagging”, “online”, “live” (infrastructure). This textual evidence independently confirms Pace of Delivery and Platform as the top two concerns identified through statistics and modeling.

Limitation: the lexicon-based sentiment tool was found to systematically misclassify polite, hedged criticism as positive — all 132 comments containing the word “rushed” (e.g., “wish there was more time per topic, felt rushed”) were scored as Positive sentiment despite describing a clear pacing complaint. This means the reported 10% Negative sentiment figure likely understates true dissatisfaction, and keyword-level analysis should be treated as a necessary complement to, not a replacement for, sentiment scores.

	Open_feedback	Sentiment
0	Session quality varied depending on the trainer.	Neutral
1	Would like the syllabus to be updated more fre...	Positive
2	Loved the hands-on projects, they really built...	Positive
3	Average experience, some sessions were better ...	Positive
4	Faced frequent audio/video issues during classes.	Neutral
5	Well-paced sessions and very approachable trai...	Neutral
6	The portal for assignments was confusing and s...	Negative
7	The portal for assignments was confusing and s...	Negative
8	Very satisfied, the mentors were patient and e...	Positive
9	Session quality varied depending on the trainer.	Neutral

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Sentiment
Positive      1083
Neutral       665
Negative       201
No Feedback     51
Name: count, dtype: int64
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5. Scope of Improvement — Ranked List

Consolidating evidence from all four analytical methods, the following ranked list identifies specific improvement areas, the evidence behind each, the segment most affected, and a suggested priority for management action.

Issue	Supporting Evidence	Affected Segment	Priority
Pace of Delivery is too fast	Lowest overall mean (3.52/5) with high variance (SD 0.94); 3rd-strongest predictor of low satisfaction in regression model (coeff. -1.24); 132 open-ended comments explicitly mention feeling “rushed”	Most pronounced in the Full Stack Development program (mean 3.18) and in Feb–Mar 2026 batches	High
Platform / Infrastructure issues	Lowest overall mean (3.55/5) and highest variance of all dimensions (SD 0.96); negative-sentiment keyword analysis surfaced “lagging”, “platform”, “online”, “live”	Concentrated in live/online delivery mode; weakest in Full Stack Development and Competitive Programming programs	High
Curriculum pacing vs. depth mismatch	Curriculum_relevance has the weakest regression coefficient (-1.21) among the five dimensions; text mining shows overlap between pace complaints and difficulty with newly introduced concepts (“hard”, “struggled”, “introduced”, “new”)	General, but co-occurs with the same programs/batches flagged for pace issues	Medium
Support & Mentorship — protect current standard	Second-strongest predictor of satisfaction (coeff. -1.53) and second-highest correlation with Overall Satisfaction ($r=0.73$); already a comparative strength (mean 3.98, lowest variance)	Org-wide; a small “Dissatisfied” student cluster (lowest across all dimensions) still needs targeted outreach	Medium
Trainer Effectiveness — maintain as a core strength	Strongest single predictor of Overall Satisfaction in both correlation ($r=0.78$) and regression (coeff. -2.35); already highest mean (3.95) with lowest variance	Org-wide strength; no corrective action needed, protect current trainer quality standards	Low

6. Recommendations

- Review and adjust pacing in the Full Stack Development curriculum specifically, and in batches scheduled from February 2026 onward, where pace-related dissatisfaction is most concentrated.
- Audit platform/infrastructure reliability for live and online delivery formats; lag and access issues are a recurring, specific complaint distinct from content quality.
- Introduce a structured check-in or early-warning process for students matching the “Dissatisfied” cluster profile (uniformly low ratings across dimensions), since this group is unlikely to be reached by dimension-specific fixes alone.
- Maintain current trainer recruitment/training standards and mentorship practices — both are validated strengths and the strongest drivers of overall satisfaction; resources should not be diverted away from these areas.

- When reviewing future feedback text, supplement sentiment scores with direct keyword/phrase searches (e.g., “rushed”, “lagging”) to avoid under-counting politely worded criticism.

7. Assumptions & Limitations

- Missing rating values (18–41 records per dimension, under 2% each) were imputed using the median, appropriate for ordinal 1–5 scale data.
- 51 records with no open-ended feedback were treated as blank and excluded from text/sentiment analysis; they were not assigned a fabricated sentiment or theme.
- “Low satisfaction” for predictive modeling was defined as an Overall Satisfaction score of 3 or below; this threshold was chosen to yield a reasonably balanced classification target (~34% positive class) and is a judgment call, not a fixed organizational standard.
- Segment-level findings (e.g., by Program or Batch) are based on varying sample sizes, ranging from 19 to 192 respondents per batch; findings from smaller segments (under ~30 respondents) are noted as lower-confidence and should be validated against future data before major decisions are based on them.
- Sentiment analysis used a lexicon-based method (VADER) as required by the project's classical-methods-only scope; as noted in Section 4, this approach under-detects politely phrased criticism, and Negative-sentiment figures should be read as a conservative lower bound.
- This report reflects feedback data as consolidated at the time of analysis; it does not incorporate real-time or streaming updates, consistent with the project's defined scope.

8. Conclusion

The convergence of four independent analytical methods — descriptive statistics, correlation, predictive modeling, and text mining — on the same two priority areas (Pace of Delivery and Platform/Infrastructure) provides strong, well-corroborated grounds for management action. At the same time, Trainer Effectiveness and Support & Mentorship stand out as validated organizational strengths that should be actively protected. Addressing the pacing and infrastructure issues identified here, particularly within the specific programs and batches flagged in Section 5, represents the most impactful and evidence-backed opportunity to improve the overall student training experience.